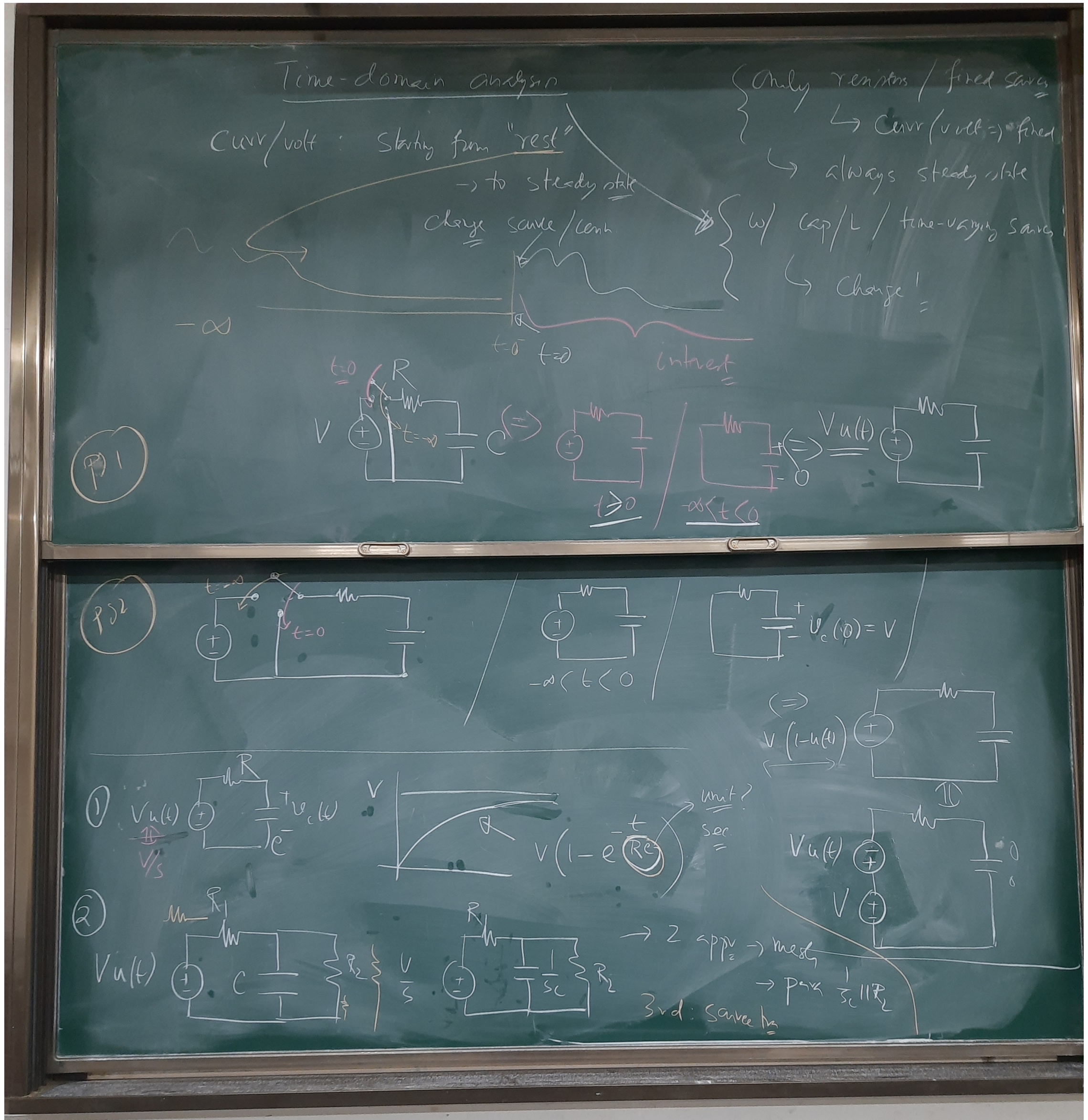


## Time-domain analysis

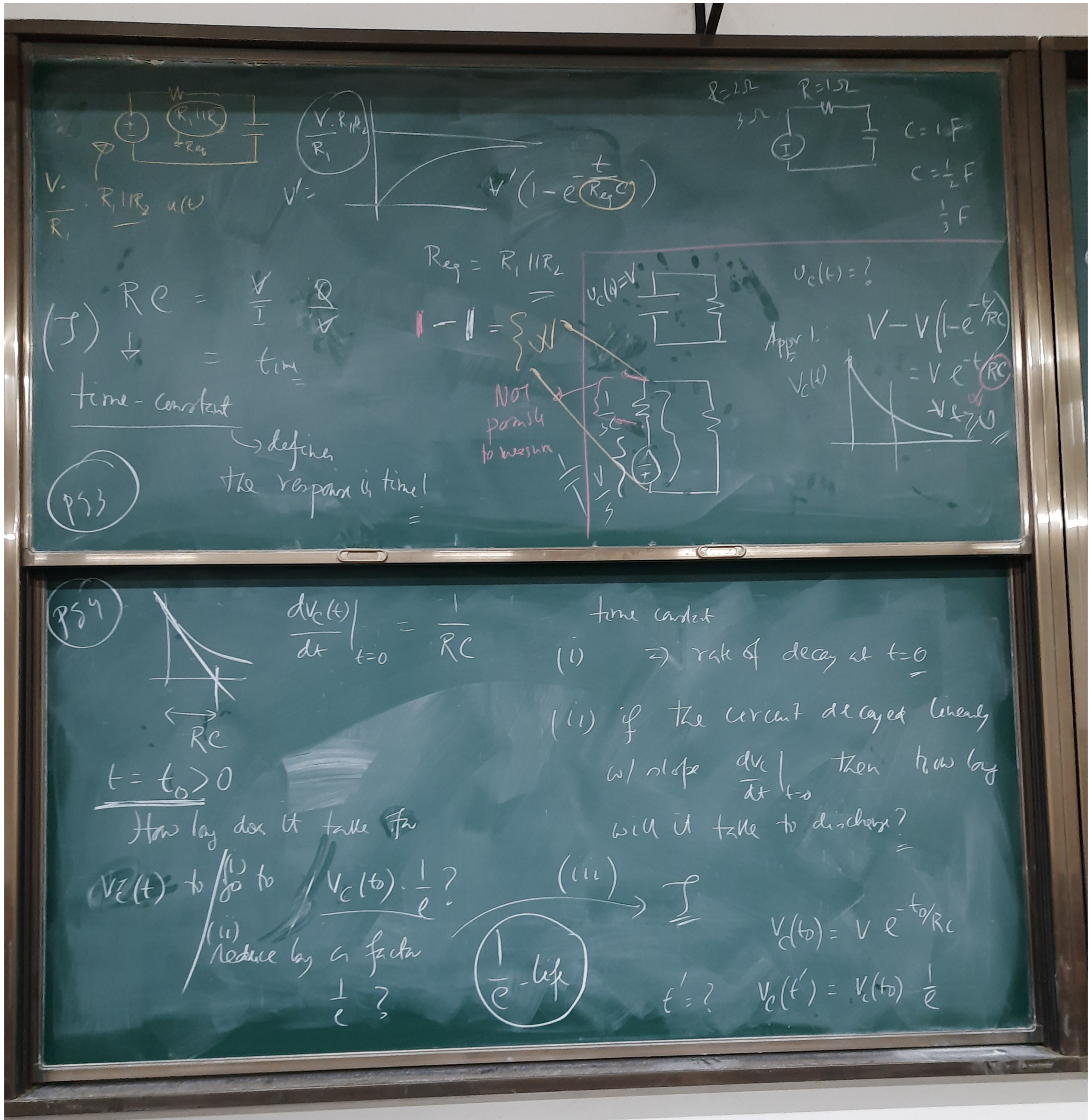
Curr/volt : Starting from "rest"
→ to steady state
Change source/conn

{ Only resistors / fixed sources
↳ Curr (volt) ⇒ fixed
↳ always steady state

{ w/ cap/L / time-varying sources
↳ Change !

→ 2 appr → mesh
→ para $\frac{1}{sC} \parallel R_2$

3rd : source tr.



$\frac{V}{R_1} \cdot R_1 \| R_2 \; u(t)$

$V' = \frac{V}{R_1} \cdot R_1 \| R_2$

$V'\left(1 - e^{-\frac{t}{R_{eq}C}}\right)$

$R_{eq} = R_1 \| R_2$

$\| - \| = \{ \mathcal{W}$

NOT permissible to measure

$(s)\; RC = \frac{V}{I} \cdot \frac{Q}{V}$

$= \text{time}$

time-constant

→ defines the response in time!

PS3

$v_C(t) = ?$

Appr 1. $V - V\left(1 - e^{-t/RC}\right)$

$= V e^{-t/RC}$

$\forall\, t \geq 0$

PS4

$\left.\frac{dv_C(t)}{dt}\right|_{t=0} = -\frac{1}{RC}$

time constant

(i) ⇒ rate of decay at t = 0

(ii) if the circuit decayed linearly w/ slope $\left.\frac{dv_C}{dt}\right|_{t=0}$ then how long will it take to discharge?

$t = t_0 > 0$

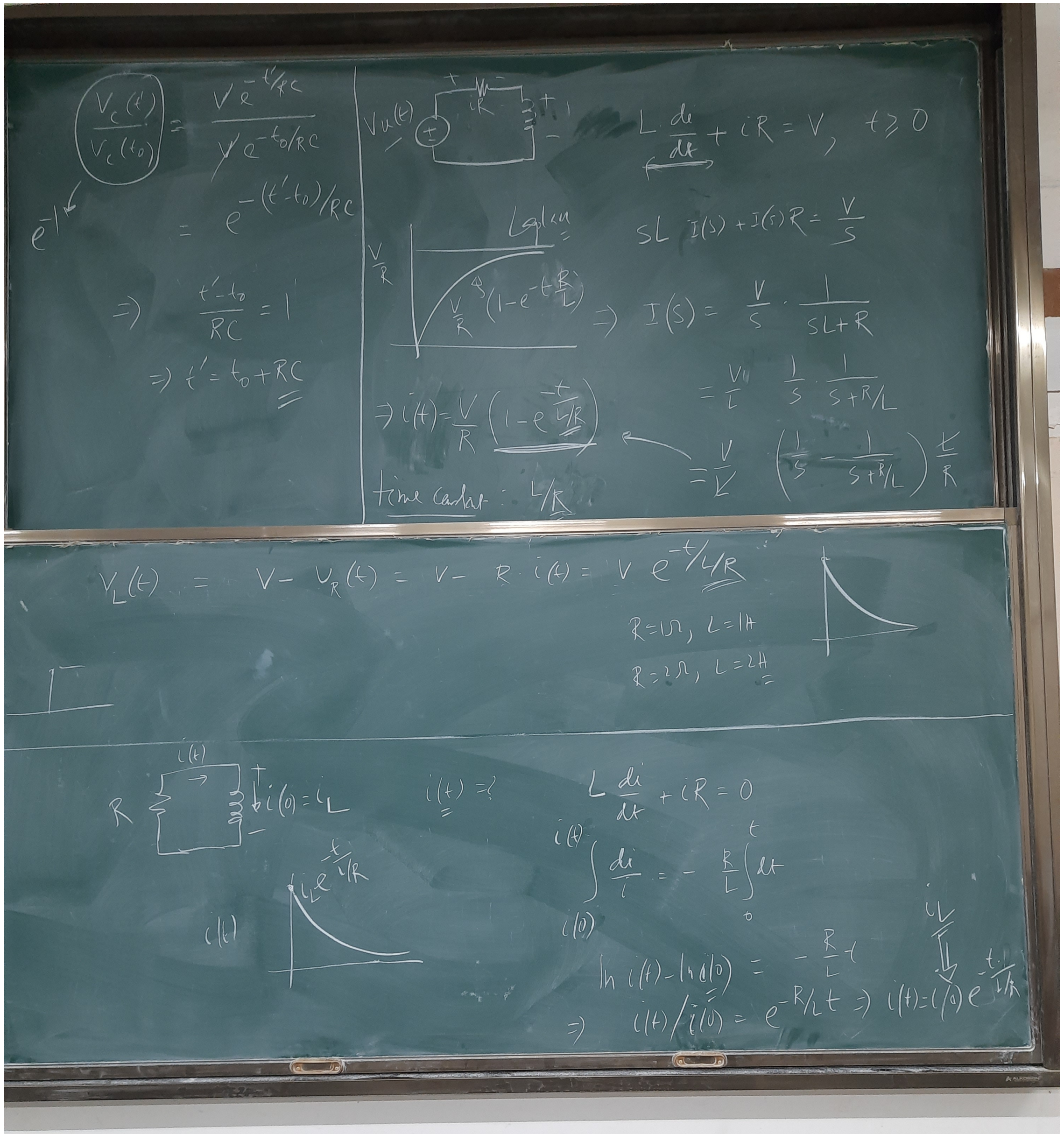
How long does it take for $V_C(t)$ to (i) go to $V_C(t_0) \cdot \frac{1}{e}$? (ii) reduce by a factor $\frac{1}{e}$?

(iii) → ∫

$\frac{1}{e}$-life

$V_C(t_0) = V e^{-t_0/RC}$

$t' = ?\quad V_C(t') = V_C(t_0)\,\frac{1}{e}$



$$\boxed{\frac{V_c(t')}{V_c(t_0)}} = \frac{\cancel{V}e^{-t'/RC}}{\cancel{V}e^{-t_0/RC}}$$

$$= e^{-(t'-t_0)/RC}$$

$e^{-1}$

$$\Rightarrow \frac{t'-t_0}{RC} = 1$$

$$\Rightarrow t' = t_0 + RC$$

$$L\frac{di}{dt} + iR = V, \quad t \geq 0$$

Laplace

$$sL\,I(s) + I(s)R = \frac{V}{s}$$

$$\Rightarrow I(s) = \frac{V}{s}\cdot\frac{1}{sL+R}$$

$$= \frac{V}{L}\,\frac{1}{s}\cdot\frac{1}{s+R/L}$$

$$= \frac{V}{L}\left(\frac{1}{s} - \frac{1}{s+R/L}\right)\frac{L}{R}$$

$$\Rightarrow i(t) = \frac{V}{R}\left(1 - e^{\frac{-t}{L/R}}\right)$$

time constant : $L/R$

$$V_L(t) = V - U_R(t) = V - R\cdot i(t) = Ve^{-t/\underline{L/R}}$$

$R = 1\Omega,\ L = 1H$

$R = 2\Omega,\ L = 2H$

$i(t) = ?$

$$L\frac{di}{dt} + iR = 0$$

$$\int_{i(0)}^{i(t)} \frac{di}{i} = -\frac{R}{L}\int_0^t dt$$

$$\ln i(t) - \ln i(0) = -\frac{R}{L}t$$

$$\Rightarrow i(t)/i(0) = e^{-R/L\,t} \Rightarrow i(t) = \overset{i_L}{\overset{\Downarrow}{i(0)}}\,e^{-\frac{t}{L/R}}$$